CHARGING IS THE MOST DANGEROUS OPERATION IN THE LIFE OF THIS MACHINE — OUTDOORS, TRAINED OPERATOR, NO IGNITION SOURCE

CHARGING SEQUENCE

1 — WELD OUT
Complete every wetted joint. No threaded or elastomer joint remains.

2 — PROOF PRESSURE — 1.5 x stagnation
Hydrostatic 21 bar, 5 min. Drain, dry with warm dry nitrogen.

3 — LEAK TEST — < 1e-8 Pa.m3/s
Helium mass spectrometer under vacuum on the whole envelope.

4 — BAKE-OUT — carbon outgasses
200 C / 8 h at < 1 Pa through the charge tube, pumping throughout.

5 — COOL UNDER VACUUM — no air, ever
Let the bed reach ambient still under the pump. Do not vent.

6 — CHARGE — by mass, not volume
Admit 4.9 kg anhydrous methanol by WEIGHT from a sealed burette.

7 — DEGAS — removes dissolved air
Freeze-pump-thaw x3 on the receiver. Final pump to < 50 Pa at 0 C.

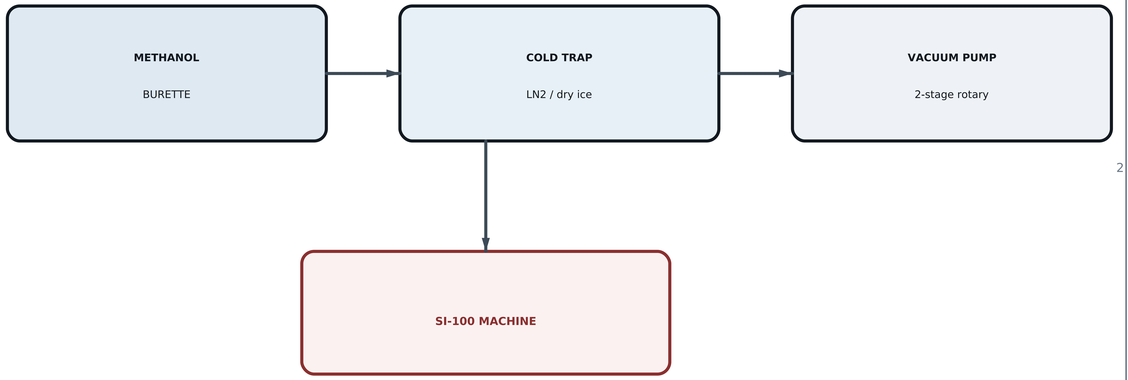
8 — PINCH-OFF — permanent
Cold-weld the charge tube, cut, then TIG cap the stub.

9 — LEAK TEST 2 — < 1e-8 Pa.m3/s
Helium bomb 3 bar / 4 h, then mass spectrometer.

10 — FIRST CYCLE — commissioning
One full solar cycle under observation. Record bed and receiver temperatures.

CHARGING RIG

SCHEMATIC



The cold trap protects the pump and, more importantly, stops methanol

vapour reaching the exhaust. Vent the pump outdoors, downwind.

CHARGING PARAMETERS

PARAMETER	VALUE	LIMIT
Methanol charge	**4.91 kg**	**by weight, ±1 %**
Purity	anhydrous > 99.8 %	water poisons the carbon
Bake-out	200 C / 8 h	< 1 Pa
Final vacuum before seal	< 50 Pa	at 0 C
Proof pressure	21 bar	1.5 x stagnation
Leak rate	**< 1e-8 Pa.m3/s**	**helium, twice**
Seal	cold-weld + TIG cap	no valve, no port

PROHIBITED

- Venting the envelope to air at any point after bake-out. Air in the bed is non-condensable gas and it never comes out again.
- Charging by volume. Methanol expands 1.2 % per 10 K; the charge is a mass specification.
- Any valve, gauge port or sight glass left on the wetted envelope. Every one is a leak path for the life of the machine.
- Charging indoors, or anywhere without free ventilation and no ignition source within 5 m.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: VACUUM AND CHARGING 4.9 kg METHANOL / ONE-TIME PERMANENT CHARGE / NO FIELD SERVICE			SCALE: NONE THIRD ANGLE PROJECTION	UNITS: mm
DRAWING No.: SI-200			REV: A	SHEET: 5 / 6
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-15